Supplementary Material

Degradation-Aware Battery Dispatch via Hybrid-Neural Network and Projected Gradient Descent

Pengfei He, Daniyaer Paizulamu, Kaidi Huang, Yufei Xi, and Lin Cheng

This document provides the proofs of Theorems 1–2, supplementary formulations for methods M1–M5, and numerical constants used in the main paper. Main-text equation numbers refer to the accompanying manuscript; equations in this document carry section prefixes. Powers are AC-side quantities, positive for discharge and negative for charge. Capacity loss and SOC are expressed in per unit (pu).

Section A: Proofs of Theorems 1 and 2

A.1 Proof of Theorem 1: convexity and possible discontinuity

Fix the day-start states, network parameters, and operating pattern $H \in \{-1, 0, 1\}^N$. Assume $E_d > 0$ and $\eta_{\text{ch}}, \eta_{\text{dis}} > 0$. Define

$$\rho_j = \begin{cases} \eta_{\text{ch}}, & H_j = -1, \\ 1/\eta_{\text{dis}}, & H_j = +1, \\ 1, & H_j = 0. \end{cases} \tag{A.1}$$

On the sign-restricted set in main-text Eq. (34), the SOC recursion becomes

$$\widehat{s}_q(\mathbf{p}) = s_0 - \frac{\Delta t_h}{E_d} \sum_{j=0}^{q-1} \rho_j p_j, \qquad q = 1, \ldots, N. \tag{A.2}$$

The idle coordinates vanish, so their choice of ρ_j has no effect. Every SOC constraint is therefore an affine inequality in $\mathbf{p}$. The power bounds and sign constraints are linear inequalities, and $p_{\mathcal{I}} = 0$ is an affine equality. Their intersection C_H is a closed convex polytope, bounded by the power limits. This proves convexity.

For the second claim, it suffices to exhibit a zero-power point with a discontinuous aging response. Write the half-cycle squared-loss contribution in main-text Eq. (17) as

$$\Phi(d, \tau) = \frac{d}{20000} \left[(a_{\text{cyc}} d/\tau + b_{\text{cyc}}) L(d) \right]^2, \quad L(d) = u_{\text{cyc}} (d - d_{\text{ref}})^3 + v_{\text{cyc}}. \tag{A.3}$$

For $d, \tau > 0$, the coefficients in Table C.2 give $a_{\text{cyc}} > 0$, $b_{\text{cyc}} > 0$, and $L(d) > 0$ on the physical depth range. Consequently,

$$\frac{\partial \Phi}{\partial \tau} = -\frac{a_{\text{cyc}} d^2 L(d)^2}{10000 \tau^2} \left(a_{\text{cyc}} d/\tau + b_{\text{cyc}} \right) < 0. \tag{A.4}$$

Consider an idle interval inside an otherwise same-direction half-cycle that is subsequently settled at a reversal within the day. Let its depth and active duration be $d > 0$ and $\tau > 0$. Strict idle adds neither depth nor active duration. Replacing the idle power by a same-direction power that produces SOC movement $\varepsilon > 0$ changes these quantities to $(d + \varepsilon, \tau + \Delta t_h)$. The full interval contributes to active duration however small its nonzero power. With sufficient SOC headroom this perturbation is feasible for all sufficiently small ε, and the corresponding power tends to zero. Yet

$$\lim_{\varepsilon \downarrow 0} \Phi(d + \varepsilon, \tau + \Delta t_h) = \Phi(d, \tau + \Delta t_h) < \Phi(d, \tau). \tag{A.5}$$

The other half-cycle depths can remain unchanged; a constant SOC offset after the perturbed interval does not change their depths. Since $x \mapsto \sqrt{q^2 + x}$ is strictly increasing, the daily cyclic-loss increment inherits the finite jump.

This cyclic jump gives a discontinuity of the total $f_\theta = \widehat{D}_{\text{cal}} + \widehat{D}_{\text{cyc}}$ whenever the calendar component is continuous along the perturbation. Such cases are admitted by the model: for example, a continuous heat map with no jump at zero (including a zero heat map) gives continuous temperatures, SOCs, and calendar loss. Thus f_θ *can* be discontinuous in the full power space. The claim does not assert discontinuity at every idle coordinate or rule out an exceptional cancellation between components. □

The closed set C_H includes boundaries where an active power reaches zero. Differentiability is asserted only locally on a strict-sign branch, $H_j p_j > 0$ for $j \in \mathcal{A}$, with the half-cycle partition fixed and positive active durations. The zero extension of an active-subspace gradient on $\mathcal{I}$ is an algebraic convention, not a claim that ordinary derivatives exist at idle coordinates.

A.2 Proof of Theorem 2: proximal–projection equivalence

Let $\mathbf{p} = \mathbf{p}^{(k)} \in C_H$, let $\mathcal{S}$ be a linear subspace, and define

$$\mathcal{K}_{\mathcal{S}} = C_H \cap (\mathbf{p} + \mathcal{S}), \qquad \Pi^G_{\mathcal{K}_{\mathcal{S}}}(\mathbf{z}) = \arg\min_{\mathbf{y} \in \mathcal{K}_{\mathcal{S}}} \|\mathbf{y} - \mathbf{z}\|^2_G. \tag{A.6}$$

Here G is the diagonal matrix in main-text Eq. (35). Positive efficiencies imply $G \succ 0$. The feasible set $\mathcal{K}_{\mathcal{S}}$ is closed and convex and contains $\mathbf{p}$. Thus the strictly convex, coercive squared-distance objective has a unique minimizer.

For any fixed vector g and $\alpha > 0$, completing the square gives

$$g^T\mathbf{d} + \frac{\mathbf{d}^T G \mathbf{d}}{2\alpha} = \frac{1}{2\alpha}\|\mathbf{d} + \alpha G^{-1} g\|^2_G - \frac{\alpha}{2} g^T G^{-1} g. \tag{A.7}$$

The last term is independent of $\mathbf{d}$. Substituting $\mathbf{y} = \mathbf{p} + \mathbf{d}$ into the proximal subproblem in main-text Eq. (36a) therefore yields

$$\mathbf{p} + \mathbf{d}^\star = \Pi^G_{\mathcal{K}_{\mathcal{S}}}(\mathbf{p} - \alpha G^{-1} g). \tag{A.8}$$

For the full active subspace $\mathcal{S} = \{\mathbf{d} : d_{\mathcal{I}} = 0\}$, every $\mathbf{y} \in C_H$ already satisfies $\mathbf{y} - \mathbf{p} \in \mathcal{S}$. Hence $\mathcal{K}_{\mathcal{S}} = C_H$, giving main-text Eq. (37). The identity requires neither convexity nor global continuity of F; in the algorithm, g is the current branch gradient extended by zeros on the idle coordinates. □

For completeness, take $H_i = H_j \neq 0$, $g_i \neq g_j$, and the oriented pair direction $\mathbf{v} = -\operatorname{sign}(g_i - g_j)(\mathbf{e}_i - \mathbf{e}_j)$. Set $a = -g^T\mathbf{v} = |g_i - g_j| > 0$ and $\gamma = \mathbf{v}^T G \mathbf{v} = G_{ii} + G_{jj} > 0$. The feasible scalar interval contains zero. Its negative portion cannot minimize $-a\delta + \gamma\delta^2/(2\alpha)$, because there the value is positive whereas it is zero at $\delta = 0$. Therefore, with $\Delta = \max\{\delta \geq 0 : \mathbf{p} + \delta\mathbf{v} \in C_H\}$,

$$\delta^\star = \arg\min_{0 \leq \delta \leq \Delta} \left[-a\delta + \frac{\gamma\delta^2}{2\alpha}\right] = \min\left\{\frac{\alpha a}{\gamma}, \Delta\right\}. \tag{A.9}$$

This establishes the one-dimensional specialization in main-text Eq. (38d). It is an identity for the quadratic subproblem and does not imply global optimality of the nonlinear dispatch objective.

Section B: Supplementary Formulations for Methods M1–M5

For schedules satisfying main-text Eqs. (32b)–(32e), define revenue and total AC energy throughput as

$$R(\mathbf{p}) = \Delta t_h \sum_{t=0}^{N-1} \pi_t p_t, \qquad Q(\mathbf{p}) = \Delta t_h \sum_{t=0}^{N-1} (p_t^{\text{ch}} + p_t^{\text{dis}}). \tag{B.1}$$

Write the minimized objective as $F_m = -R + C_m$, where C_m is the method-specific monetary penalty (EUR). A loss estimate $\widehat{D}_m$ is in pu, and its corresponding unscaled cost is $\kappa\widehat{D}_m$. These quantities are distinguished to apply the conversion factor κ only once.

B.1 M1: pure arbitrage

M1 maximizes $R(\mathbf{p})$ subject to the common dispatch constraints, with $C_1 = 0$; no aging estimate enters its objective.

B.2 M2: throughput-based cost

The linear throughput penalty is

$$C_2(\mathbf{p}) = \alpha_{\text{tp}} Q(\mathbf{p}), \qquad \alpha_{\text{tp}} = \frac{c_{\text{cell}}}{2N_{\text{cyc}}}, \qquad \widehat{D}_2 = \frac{C_2}{\kappa}. \tag{B.2}$$

Here c_{cell} is the nominal-energy-specific cell investment (EUR/kWh), whereas $C^{\text{cell}} = E_{\text{nom}} c_{\text{cell}}$ is the pack-level cell investment (EUR). N_{cyc} denotes design full-equivalent-cycle life. Both charging and discharging contribute to Q, accounting for the factor two. Calendar aging is omitted.

B.3 M3: piecewise-linear aging cost

For a four-hour block $\mathcal{B}_b$ containing 16 intervals and $a \in \{\text{ch}, \text{dis}\}$, define

$$e_b^a = \Delta t_h \sum_{t \in \mathcal{B}_b} p_t^a, \tag{B.3}$$

$$\widehat{q}_t^{\text{cal}} = f_{\text{cal}}^{\text{PWL}}(\widehat{s}_{t+1}; H_Q), \qquad \widehat{q}_b^{\text{cyc},a} = f_{\text{cyc}}^{\text{PWL}}(e_b^a; H_Q, E_d), \tag{B.4}$$

$$\widehat{D}_3 = \sum_t \widehat{q}_t^{\text{cal}} + \sum_b \left(\widehat{q}_b^{\text{cyc,ch}} + \widehat{q}_b^{\text{cyc,dis}} \right). \tag{B.5}$$

The offline PWL tables are selected using day-start native capacity SOH H_Q. The implemented M3 optimization applies reference-SOH scaling to the two components:

$$C_3 = \kappa \left[w_{\text{cal}} \sum_t \widehat{q}_t^{\text{cal}} + w_{\text{cyc}} \sum_b (\widehat{q}_b^{\text{cyc,ch}} + \widehat{q}_b^{\text{cyc,dis}}) \right], \tag{B.6}$$

$$w_x(H_Q) = \frac{y_{x,\max}(H_{\text{ref}})}{y_{x,\max}(H_Q)}, \qquad x \in \{\text{cal}, \text{cyc}\}. \tag{B.7}$$

Here $y_{x,\max}$ is the ordinate at the last breakpoint of the selected table. The reference selector $H_{\text{ref}} = 0.95$ maps to the table column with representative SOH 0.949375. Estimation comparisons use the unscaled $\widehat{D}_3$, or $\kappa\widehat{D}_3$ in EUR, rather than the scaled optimization penalty C_3.

B.4 M4: throughput-constrained candidate selection

Starting from a pure-arbitrage schedule $\mathbf{p}^{(0)}$, M4 repeatedly maximizes revenue under the common constraints and a tightened throughput bound:

$$Q(\mathbf{p}^{(k+1)}) \le (1 - \beta) Q(\mathbf{p}^{(k)}). \tag{B.8}$$

An external neural estimator scores complete schedules and selects

$$k^\star \in \arg\max_{k \in \mathcal{I}_{\text{eval}}} \left[R(\mathbf{p}^{(k)}) - \kappa f_\theta(\mathbf{p}^{(k)}; h_d, \mu_d) \right]. \tag{B.9}$$

Thus the neural estimate is used for candidate selection outside the revenue-maximizing subproblems.

B.5 M5: hybrid estimation and subspace PGD

M5 uses $C_5 = \kappa(\widehat{D}_{\text{cal}} + \widehat{D}_{\text{cyc}})$ and the HNN–PGD formulation in main-text Sections II–III. To make cross-day accounting explicit, define the pending half-cycle valuation

$$\mathcal{R}(q, d, \tau) = \sqrt{q^2 + \Phi(d, \tau)} - q, \qquad R_0 = \mathcal{R}(q_{\text{cyc},0}, d_0, \tau_0), \quad R_N = \mathcal{R}(q_{\text{cyc},N}, d_N, \tau_N). \tag{B.10}$$

An empty pending half-cycle has $\Phi = 0$. The complete daily cyclic ledger, consistent with the label in main-text Eq. (25), is

$$\widehat{D}_{\text{cyc}} = (q_{\text{cyc},N} - q_{\text{cyc},0}) + R_N - R_0. \tag{B.11}$$

Here $q_{\text{cyc},N} = \sqrt{q_{\text{cyc},0}^2 + \sum_{i \in C_d} \Phi(d_i, \tau_i)}$, so the square-root expression in main-text Eq. (18) gives the settled increment. Subtracting R_0 prevents double counting of inherited residual damage; signed daily ledger increments are not clipped, and residual valuation does not change native settled capacity.

For the same-direction pair search in main-text Eq. (38), select the pair maximizing $a\Delta$ and initialize $\alpha = \gamma\Delta/a$. Recompute gradients between updates and halve rejected step sizes. Normalize $|p_i| < \epsilon_P$ to zero before evaluation. A trial $\mathbf{y}$ is accepted only if it retains H, satisfies $g^T(\mathbf{y} - \mathbf{p}^{(k)}) < 0$, and

$$F(\mathbf{p}^{(k)}) - F(\mathbf{y}) \ge \max\{\epsilon, -c_A g^T(\mathbf{y} - \mathbf{p}^{(k)})\}. \tag{B.12}$$

Zero-boundary trials are backtracked. Initial branches use the pure-arbitrage schedule and nine throughput-capped schedules with $Q(\mathbf{p}) \le rQ_{\text{M}}$, $r = 0.1, \ldots, 0.9$; these caps apply only to initialization. A branch stops when the decrease over a full PGD–pair round is less than ϵ.

Section C: Key Parameter Values

In the location column, *Main* denotes an equation in the manuscript. Physical and aging constants follow the Sony LFP/Naumann model in SimSES 2.1.1 and the saved HNN parameters. The lifecycle power limits correspond to the runs used in the paper's comparison figures; the training trajectories use a separate AC range of $[-9.6, 9.6]$ kW. Displayed derived values and normalization statistics are rounded to nine significant digits.

Table C.1: Scheduling and thermal parameters.

Symbol	Meaning and equation location	Value	Unit
N	Intervals per day; Main (14), (29), (32)	96	–
Δt_h	Dispatch interval; Main (5), (32d)	0.25	h
Δt_s	Dispatch interval in seconds; Main (8)	900	s
δt_s	Fine integration step; Main (7), (14)	10	s
m	Fine steps per interval; Main (8)	90	–
E_{nom}	Nominal pack energy; Main (4)	9.6	kWh
η_{ch}	Planning SOC charge coefficient; Main (5), (35)	0.91	–
η_{dis}	Planning SOC discharge coefficient; Main (5), (35)	0.97	–
$P_{\text{ch}}^{\max}$	Lifecycle AC charge limit; Main (32b), (34)	10	kW
$P_{\text{dis}}^{\max}$	Lifecycle AC discharge limit; Main (32b), (34)	9.216	kW
$\underline{s}, \overline{s}$	Planning SOC bounds; Main (32e), (34)	0.1, 0.9	pu
T_a	Ambient temperature; Main (7), (12)	25	°C
R_{th}	Pack thermal resistance; Main (7)–(9)	0.0228301873	K/W
C_{th}	Pack heat capacity; Main (7), (8)	70070	J/K
α_T	Fine-step retention; Main (8)	0.993748872	–
A	Interval retention; Main (9), (12)	0.568719833	–
B	Equivalent-heat gain; Main (9), (12), (27)	0.00984620699	K/W

Table C.2: Calendar- and cyclic-aging coefficients.

Symbol	Meaning and equation location	Value	Unit
k_{ref}	Calendar reference coefficient; Main (13)	1.2571×10^{-5}	pu $\text{s}^{-1/2}$
E_a	Activation energy; Main (13)	17126	J/mol
R_g	Gas constant; Main (13)	8.3144598	J/(mol K)
T_{ref}	Reference absolute temperature; Main (13)	298.15	K
a_{cal}	SOC stress coefficient; Main (13)	2.8575	–
b_{cal}	SOC stress offset; Main (13)	0.60225	–
s_{ref}	Reference SOC; Main (13)	0.5	pu
a_{cyc}	Rate coefficient; Main (16)	0.0630	h %/$\sqrt{\text{FEC}}$
b_{cyc}	Rate offset; Main (16)	0.0971	%/$\sqrt{\text{FEC}}$
u_{cyc}	Depth stress coefficient; Main (16)	4.0253	–
v_{cyc}	Depth stress offset; Main (16)	1.0923	–
d_{ref}	Reference cycle depth; Main (16)	0.6	pu

The lifecycle AC bounds follow from a 9.6 kW DC limit and native converter efficiencies of 0.96 in each direction: $P_{\text{ch}}^{\max} = 9.6/0.96$ and $P_{\text{dis}}^{\max} = 9.6 \times 0.96$. These converter efficiencies differ from the planning SOC coefficients in Table C.1.

In the Arrhenius factor of main-text Eq. (13), T and T_{ref} must be in kelvin: $T[\text{K}] = T[^\circ\text{C}] + 273.15$. Thermal states and NN temperature inputs are stored in degrees Celsius; temperature differences and scales are in kelvin. In the cyclic law, s, d are dimensionless fractions, $c = d/\tau$ has unit h^{-1}, and FEC denotes full-equivalent-cycle count. The coefficient $k(d, c)$ uses percentage points per square root of FEC, which explains its division by 100 in main-text Eq. (17).

Table C.3: Frozen training-set statistics and loss normalization.

Symbol	Meaning and equation location	Value	Unit
$m_{x,1}$	Power mean; Main (20)	−0.184358425	kW
$m_{x,2}$	SOC mean; Main (20)	0.505740947	pu
$m_{x,3}$	Temperature mean; Main (20)	26.1717141	°C
$m_{x,4}$	Calendar-loss mean; Main (20)	0.0678168748	pu
$m_{x,5}$	Cyclic-loss mean; Main (20)	0.0318777447	pu
$r_{x,1}$	Power scale; Main (20)	3.47864706	kW
$r_{x,2}$	SOC scale; Main (20)	0.241131151	pu
$r_{x,3}$	Temperature scale; Main (20)	1.35127907	K
$r_{x,4}$	Calendar-loss scale; Main (20)	0.0444703449	pu
$r_{x,5}$	Cyclic-loss scale; Main (20)	0.0247761339	pu
m_Q	Heat-output mean; Main (23)	53.7506225	W
r_Q	Heat-output/loss scale; Main (23), (28)	92.4172786	W
r_T	Temperature-loss scale; Main (29)	1.23741683	K
ϵ_{cal}	Calendar-loss scale; Main (30)	10^{-6}	pu
λ_Q	Weight of $\mathcal{L}_Q$ in stage II; Main (31)	0.1	–

The component order is $x = (p, s, T, q_{\text{cal},0}, q_{\text{cyc},0})$. The r_x entries are population standard deviations fitted to active training intervals; all statistics are frozen after fitting. The heat network has layer widths 5–8–8–1, with tanh hidden layers and a linear output (129 trainable parameters). The temperature and calendar-loss terms in main-text Eq. (31) each have unit weight.

Table C.4: Method-specific constants.

Symbol	Meaning and equation location	Value	Unit
N_{cyc}	M2 design cycle life; Eq. (B.2)	6000	FEC
α_{tp}	M2 throughput cost; Eq. (B.2)	0.01125	EUR/kWh
Δt_{block}	M3 throughput block; Eq. (B.3)	4 (16 intervals)	h
H_{ref}	M3 reference-SOH selector; Eq. (B.7)	0.95	pu
β	M4 throughput contraction; Eq. (B.8)	0.03	–
r	M5 initial throughput fractions; Main III.B	$0.1, 0.2, \ldots, 0.9$	–
ϵ_P	M5 zero-power tolerance; Section B.5	10^{-6}	kW
c_A	M5 sufficient-decrease factor; Eq. (B.12)	10^{-4}	–
ϵ	M5 acceptance/round-stop threshold; Eq. (B.12), Main Algorithm 1	10^{-7}	EUR

Economic parameters

The costs below describe a fixed engineering scenario. The installed investment is split into cells (45%), power conversion system (PCS) and other equipment (25%), and engineering, procurement, and construction (EPC, 30%). Define

$$C^{\text{sys}} = C^{\text{cell}} + C^{\text{PCS}} + C^{\text{EPC}}, \qquad C^{x} = E_{\text{nom}} c_{x}, \quad x \in \{\text{sys}, \text{cell}, \text{PCS}, \text{EPC}\}, \tag{C.1}$$

$$\kappa = \frac{C^{\text{cell}}}{1 - \text{SOH}^{\text{EoL}}} = \frac{1296}{1 - 0.8} = 6480 \text{ EUR/pu}. \tag{C.2}$$

The cell investment remains fixed at its nominal value; it is not multiplied by the evolving SOH.

Table C.5: Lifecycle economic parameters for the 9.6 kWh pack.

Symbol	Meaning and equation location	Value	Unit
c_{sys}	Total installed unit cost; Eq. (C.1)	300	EUR/kWh
C^{sys}	Total installed investment (100%); Eq. (C.1)	2880	EUR
c_{cell}	Cell unit cost; Eqs. (B.2), (C.1)	135	EUR/kWh
C^{cell}	Cell investment (45%); Eq. (C.2), Main (32a)	1296	EUR
c_{PCS}	PCS/other equipment unit cost; Eq. (C.1)	75	EUR/kWh
C^{PCS}	PCS/other equipment investment (25%); Eq. (C.1)	720	EUR
c_{EPC}	EPC unit cost; Eq. (C.1)	90	EUR/kWh
C^{EPC}	EPC investment (30%); Eq. (C.1)	864	EUR
SOH^{EoL}	Native capacity retirement threshold; Eq. (C.2), Main IV.A	0.8	pu
κ	Capacity-loss cost coefficient; Eq. (C.2), Main (32a)	6480	EUR/pu

The throughput coefficient in M2 follows from the same cost basis:

$$\alpha_{\text{tp}} = \frac{135}{2 \times 6000} = 0.01125 \text{ EUR/kWh} = \frac{C^{\text{cell}}}{2 N_{\text{cyc}} E_{\text{nom}}}. \tag{C.3}$$

The degradation penalty uses cell investment, while cash-recovery accounting uses the total installed investment. Electricity prices are time-varying inputs in EUR/kWh, rather than constants in these tables. The downloaded Germany/Luxembourg day-ahead series is used as a perfect-foresight deterministic arbitrage benchmark. Day-start states (h_d, μ_d), available energy E_d, gradients, and search step sizes likewise vary with the day or iterate and are not fixed constants.